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Automatic generation of second level space boundary geometry from IFC models

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Abstract

Building Information Modeling aims at the use of digital building data in all planning processes. However, several challenges arise when using BIM data in the IFC file format to create building simulation models. Complex, detailed or erroneous data and the lack of software interfaces complicate and delay the simulation preprocessing. One way to reduce the simulation expert's effort is to provide space boundary data, which can be applied in both CFD and BEPS. This paper presents an algorithm for the generation of space boundaries based on the IFC data model. The focus is on model error resistance and simultaneous support of CFD and BEPS to accelerate simulation preprocessing, avoid redundant geometric model setup, and simplify the simulation result exchange.

Key innovations

- Introduction of an automated preprocessing algorithm generating space boundaries
- Application in the setup of simulation models for CFD and BEPS
- Reduction of the required preparation time of a simulation model and removal of modeling errors

Practical implications

This paper introduces an algorithm for space boundary generation based on IFC building models. The proposed method simplifies and accelerates the application of IFC data in numerical building simulation software.

Introduction

Building Information Modeling (BIM) refers to a cooperative working methodology based on digital data that contains all information relevant to a building's life cycle. An essential component of BIM is the 3D building model, which is continuously enriched with additional or updated information as the planning process progresses. Typical planning information includes geometries and properties of architectural and technical building components as well as time, cost, and maintenance planning data. To enable the exchange of building data among project participants, the international organization buildingSMART has developed the open data model Industry Foundation Classes (IFC) (ISO 16739-1:2018).

In numerical building simulations, the use of existing digital data can contribute to improved and simplified processes. For example, time-consuming geometric remodeling could be avoided. However, since the data is usually not designed and exported simulation-specific, extensive preparation steps may have to be carried out to eliminate modeling and export errors. To automate the model preparation, various programs have been developed in academic research. Some of them are out-dated or no longer accessible.

The algorithm presented in this paper is part of a toolchain, which aims to provide the basic methods to derive simulation models from digital building data. Therefore, the proposed algorithm generates Space Boundaries (SBs) based on IFC files for the application in Computational Fluid Dynamics (CFD) and Building Energy Performance Simulation (BEPS).

The paper first gives a brief overview of SBs, their generation, and challenges in simulation applications. Subsequently, the requirements for the simulation and the SB generation algorithm are explained. Finally, the results are shown and applied in simulations using an academic IFC example model.

Space Boundary Definition

SBs are the boundary surfaces of spaces and the interfaces of building elements in contact with air. In building-related CFD, SBs are usually required to investigate the air volume for temperatures, flow velocities, and gas concentrations by imposing boundary conditions on them. In BEPS, SBs are the heat transfer surfaces between zones of different temperatures. A distinction is made between first level and second level SBs, see figure 1. At the first level, SBs are pure space delimiters. At the second level, SBs take changes in building elements or spaces on the other side into account, which makes them well suited for the application in BEPS. Regardless of the level, SBs cannot be explicitly extracted from a purely geometric IFC model.

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Within the IFC standard in version 4.1, published by buildingSMART in 2018, SBs are represented by the class IfcRelSpaceBoundary. In addition to the geometric representation and the assignment to a space and a building element, SBs can be characterized by their boundary type. For instance, second level SBs can be internal or external, physical or virtual, and of type 2a or 2b. While 2a SBs allow heat exchange between two spaces, 2b SBs are in front of a heat blocking building element (figure 1).

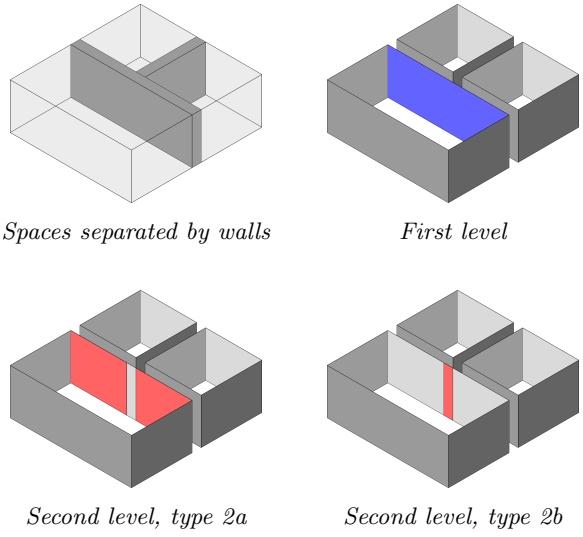


Figure 1: Distinction between first and second level SBs (based on buildingSMART (2018)).

State of the art

Challenges using IFC data

According to the authors' research, the use of 3D geometries from IFC files in CFD has only recently been applied in industry. Among other reasons, that is because IFC files are not provided and software to handle them is unknown or unavailable. Meanwhile, an increasing number of software providers are creating IFC import interfaces in their tools. These include, for example, the commercial programs BIMHVACTool, a graphical preprocessor for OpenFOAM (Weller et al., 1998), and BIM inside ANSYS, an interface for the ANSYS simulation environment. Besides, there are free and partially web-based services for converting IFC geometries into various geometry data formats such as Standard Triangle Language (STL) and Wavefront OBJ, for example IfcConvert and BIMvision.

The use of IFC files can be challenging with or without commercial software, especially for huge models with unnecessary objects, corrupted geometries, and a high level of detail. Also, modeling issues and export errors such as overlapping objects or gaps between objects can cause difficulty. Moreover, when converting the IFC into a geometry format, BIM data such as object attributes, inheritance information, and relations of objects to other entities is lost. Hence, the motivation of the proposed algorithm is to automate parts of the preprocessing steps that have to be conducted for CFD mesh creation.

Multiple BEPS tools provide an interface to IFC. Nevertheless, according to the buildingSMART International Standards Implementation Database (2020), the majority of them only supports the prior version IFC2x3. This concerns, for example, the commercial tools Simergy (2020) and IDA ICE as well as the BimServer plugin OsmSerializer (2016) used in the open source software OpenStudio (2020). The EnergyPlus supporting middleware SimModel integrated into Simergy (O'Donnell et al., 2011; Nytsch-Geusen et al., 2019) and the toolchain BIM2Modelica (Nytsch-Geusen et al., 2019) are restricted to IFC2x3, too. According to Mediavilla et al. (2018), the RenoBIM software, a preprocessor for EnergyPlus developed within the BERTIM (2019) project, is capable of reading IFC4 data. The toolchain suggested by the BIM2SIM project supports EnergyPlus simulation based on IFC4 files. For this software, the availability of SBs is crucial as their representation is used as geometry input for the Building Energy Model (BEM). In the workflow proposed by Andriamamonjy et al. (2018), SBs must be defined as well.

Despite SBs are defined within the IFC standard, several geometric and non-geometric issues can arise when using SBs from IFC files in BEPS. Issues can be caused by incorrect export settings, insufficient IFC support in the authoring tool, and modeling that was not adapted for BEPS. Typical geometric errors are missing, duplicate, deformed, incorrectly oriented, or overlapping SBs, that are not enclosing a space correctly. Also, SBs with missing or wrong semantic information, e. g. concerning boundary types and related elements, can occur. A detailed description of occurring issues was given by Maile et al. (2013).

Because of the potential errors mentioned, an IFC model check with a focus on SBs should be performed beforehand. However, there are only a few tools supporting this. Visual verification can be performed in IFC viewers. Also, syntactic correctness in comparison with the IFC standard can be checked. Tools that support such rule checking are the IfcCheckingTool (2020), the Solibri Model Checker and IfcDoc. The latter can also be used to implement SB-specific custom rules, as Wimmer et al. (2017), Pinheiro et al. (2018) and Ying and Lee (2017) proposed. Unfortunately, there is no comprehensive software for checking geometric and complex BEPS-specific rules. Though, Ying and Lee (2020) introduced a tool, detecting geometric errors with a focus on watertight geometry using a Monte Carlo ray tracing approach.

From the challenges described in this section, a need for algorithms and tools generating correct SBs can be derived.

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Space boundary generation

SBs can be written to IFC at two different stages in BIM. A generation can be done within the CAD software during the export of the IFC file, which is shared with the simulation expert. This would be the preferred way since all information of the parametric modeled BIM objects and their relationships among each other remains available. However, the IFC export interfaces of the CAD tools may be unreliable. In such a case, the SB generation needs to be based on the geometric IFC data. Due to the IFC schema definition and the export of the CAD software, information on the relationships between the objects is largely lost. Hence, it must be algorithmically regenerated. Another challenge for the downstream generation of the SBs is the quality of the IFC data, which is influenced by errors and the level of geometry. Several algorithms to generate SBs based on three-dimensional architectural building models were presented in academic research (van Treeck and Rank, 2007; Rose and Bazjanac, 2013; Jones et al., 2013; Ladenhauf et al., 2016; Lilis et al., 2016; Nytsch-Geusen et al., 2019). They are summarized in table 1 listing important prerequisites and methods. Most of the implementations that arose from the algorithms are not actively supported anymore, are based on prior IFC versions, or are not publicly accessible.

Typically, SB generation is done in three main steps. In the first step, the fulfillment of the semantical and geometric prerequisites of the algorithm must be ensured by preprocessing. Especially to reduce clashes between objects beforehand, Boolean operations have to be used. This is for example demonstrated in Lilis et al. (2015), proposing methods to ensure the needs of their CBIP algorithm (Lilis et al., 2016). Important requirements of the algorithms mentioned above are summarized in table 1.

In the second step, first level SB geometry is generated. For this, some approaches rely on the presence of IfcSpace geometry. If IfcSpaces are used, first level SBs are generated by intersecting the faces of spaces and building elements by Boolean operation, which requires a correct and consistent model without gaps and intersections. Algorithms that do not rely on IfcSpaces must extract faces in contact with air by other approaches. Van Treeck and Rank (2007) create an edge-face graph of the building by mutually intersecting solid objects using Boolean operations. Evaluating the graph's adjacency information allows detecting first level SBs and ensures watertight spaces. Jones et al. (2013) identify spaces by analyzing view factors between polygons calculated by ray-face intersection.

Second level SB geometry is generated in the third step. Except for van Treeck and Rank (2007), the mentioned algorithms use a projection approach. In this, polygon pairs are determined according to various criteria such as parallelism, distance, and the presence of reverse-oriented surface normals. SBs are then created by mutual projecting and clipping the paired faces. In the approach of van Treeck and Rank (2007), imprints on faces created by extruded solids describe a subset of second level SBs.

A combination and adaptation of the presented approaches has the potential to make the SB generation more error resistant and to fulfill the semantic and geometric requirements of CFD and BEPS. This can lead to acceleration of simulation preprocessing, avoidance of redundant geometric model setups and simplification of the exchange of simulation results.

Simulation demands

For a CFD simulation of a zone based on STL input files, a manifold watertight triangular surface mesh is needed, which contains only objects relevant to set boundary conditions. To simplify the meshing process, the faces should be consistent and correctly oriented, and triangulated in a way that a vertex is connected to three edges. Inner, double, and overlapping coplanar surfaces, as well as surface intersections, are not allowed. The above requirements can be satisfied by the proposed algorithm. However, remeshing of the geometry must be done in separate meshing software, for example, to eliminate faces with high perimeter to area ratio or faces with very acute angles. Also, a collision of one or more correct geometries in a vertex or edge can cause problems when generating a surface mesh, even though it is geometrically valid. Such cases must be solved either manually or by a wrapping algorithm in the meshing software.

For BEPS, the requirements of the IFC standard must be met in particular. Additional requirements must also be fulfilled:

- SBs form a closed shell around the space
- SB normals pointing outward of the space
- Spaces and SBs shall not intersect each other
- Simplifying container elements (IfcCurtainWall)
- Link to IfcExternalSpatialElement for facade SBs
- The SBs of a void filling object are coplanar to the SBs of the parent element

Methods

The proposed algorithm aims at generating SBs that meet both, the stated requirements for CFD and for surface-based IFC SBs. To be independent of data and model correctness, IfcSpace geometry should not be used for first level SB generation. For this reason and to generate watertight adjacent first level SBs, the approach of van Treeck and Rank (2007) is suitable. However, it is adapted to a purely surface-based approach. Thus, no closed BREP solids with correct normals are assumed, which are necessary in

Table 1: Space boundary generation algorithms. References to columns from left to right: van Treeck and Rank (2007), Jones et al. (2013), Rose and Bazjanac (2013), Ladenhauf et al. (2016), Lilis et al. (2016)

	Tre07	Jon13	Ros13	Lad16	Lil16	Proposed
Prerequisites for algorithm						
IfcSpaces for 1st level SBs			×	×	×	
Absence of clashes/gaps between space and construction			×	×	×	
Absence of clashes/gaps between constructions		×	×		×	
Watertight solid BREPs	×		×	(×)	×	
Correct surface normals	×		×	×	×	
Geometric representation	solid	polygon	polygon	polygon	polygon	all faces
Method for 1st level SB generation						
Clipping of space and construction faces			×	×	×	
Detection by view factors		×				
Evaluating edge-face graph created by Bool. operation	×					×
Method for 2nd level SB generation						
Projection and clipping of faces		×	×	×	×	×
Extension of solids/faces before 1st level SB generation	×					×
Could generate geometry and properties of IFC4 SBs		×	×	×	×	×

other algorithms, for example, to eliminate collisions between components. Furthermore, virtual faces or incorrectly modeled objects, e. g. unrealistically thin objects, can be handled. Based on the extracted first level SBs, projection and clipping allow the generation of all second level subdivisions required by IFC. Ray tracing operations ensure the distinction of type 2a and 2b faces and recognition of corresponding faces, to address changes in material, building element or space. The algorithm involves several steps:

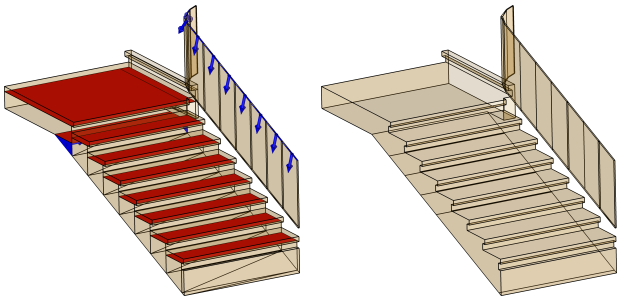
1) Parsing IFC data and conversion to BREP
2) Geometric healing (partially optional)
3) Geometric simplification (partially optional)
4) Creation of face-only data structure
5) Offsetting faces
6) Creation of edge-face graph by Boolean operation
7) Traversing graph to identify e. g. first level SBs
8) Export for CFD as STL (optional)
9) Projection and clipping of faces
10) Identification of SB properties by ray tracing
11) Merging of SBs with same properties
12) Export for BEPS as IFC

The algorithm is currently being implemented in C++. The IFC data is parsed and geometrically interpreted using IfcOpenShell (2020), a software library supporting IFC's geometric resources. It is based on the open source 3D modeling kernel Open Cascade Technology (OCCT, 2020), which is later used for the geometric operations of the presented algorithm. Using IfcOpenShell, OCCT shapes are created for the three-dimensional building elements of relevant IFC classes. The shape geometry is stored as BREP, with no restrictions on the curvature of edges and faces.

Procedure

Especially for complex objects with high levels of detail, the shapes can be faulty. The reason for this is usually an incorrect geometric description in the IFC file. A common problem is the presence of non-closed shells and inner, overlapping, or wrongly oriented faces. As long as the gaps in the hull of the shape do not exceed a critical value (a few centimeters), this kind of geometry does not affect the algorithm. Yet, to speed up later processes, an attempt is made to heal faulty shapes. For this, smaller gaps are first closed by face extension. Then, the enveloping surface of the shape is extracted according to the geometric analysis proposed by van Treeck and Rank (2007), ensuring a consistent face orientation. If possible, coplanar adjacent surfaces are unified.

The result of a healing process is exemplified in figure 2. The staircase is taken from the IFC building presented later (figure 5). It consists of a set of faces that form partially closed and partially non-closed volumes. Moreover, the staircase has inner and wrongly oriented faces as well as faces that are unnecessarily triangulated. As figure 2 (b) shows, the errors can be eliminated by the integrated healing process.

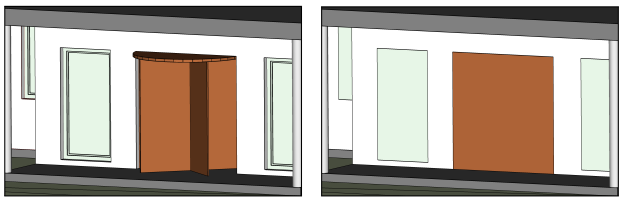


(a) before *(b) after*

Figure 2: Healing of an IfcStair's geometry. Inner surfaces are colored red, faces with incorrect normals (arrows) are shown in blue.

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Window and door objects are often described geometrically detailed, which is not practical for simulation. Therefore, the geometry of associated IfcOpeningElements is used instead. Because of the face-based approach, geometric correctness of a wall is irrelevant, as long as the usually simple opening element is correct. Thus, an opening can be expressed as two faces lying on two wall faces, reducing the geometric complexity. Assemblies of components, such as IfcCurtainWalls, are simplified by rectangular enveloping surfaces as required by BEPS. Figure 3 shows some simplifications and indicates that holes can be closed within the process, too.



(a) before *(b) after*

Figure 3: Geometric simplification of windows and a revolving door.

Regardless of healing and simplification, all surfaces are collected from the BREPs. Subsequently, the faces are offset by a few centimeters to fill gaps within and between components. Boolean intersection of the faces and adjacency evaluation results in an edge-face graph of the building, in which faces are connected to other faces via edge pairs. Faces with unconnected edges are removed. The orientation of a face's surface normal is not important for the subsequent processes. However, if the faces were part of a healed shape, the correct orientation is known.

To find first level SBs and to detect inner faces, the graph is traversed starting from a randomly selected face, collecting neighbor faces recursively. If the correct face normal of the neighbor face is known but not compatible, the search is interrupted and restarted with another initial face or another face orientation. If it is unknown, the orientation of the face is adjusted to be compatible. If a surface has several neighbors at one edge, the surface with the smaller angle is selected. The search ends with a closed space if no more new faces are found. Checks follow to verify the correctness of the found space. Spaces must not overlap, the surface normals must point into the enclosed volume, and component surfaces may only be used once. For CFD-based applications, the clustered faces are forwarded to the triangulation and export process along with information about a SB's related space and building element.

To achieve a geometric second level subdivision, faces that do not exceed a maximum distance, are mutually projected and clipped. The maximum distance is a user-defined value, which is also used to distinguish the boundary types (2a or 2b). For BEPS, attributes like boundary type, corresponding face, or material composition between the face pair need to be assigned. This is done based on a face-line-intersection (ray tracing) as depicted in figure 4. Here, lines are generated based on SB surface normals and examined for contact with other faces. The first level SB that is hit first, represents the corresponding SB. If no SB is hit within the maximum distance, a 2b boundary type is assigned. Changes in the material are derived from the order in which the faces are hit. Finally, SBs are merged with their neighbors if they share all attributes.

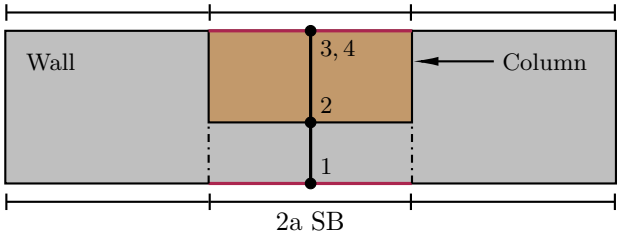


Figure 4: Face-line-intersection between two 2a SBs (red) seen from above. There are four intersection points: wall face (1), column face (2), wall and column face (3, 4).

Data export

As described in the requirements, a triangulated mesh should be created that just needs to be stitched before the surface meshing can be performed. During stitching, adjacent triangles are topologically connected via geometrically identical edges. Nonetheless, surface wrapping of the geometry is required, if additional objects are added after the export. The geometry is exported as a single STL file. The file contains STL solids, which are composed of faces belonging to the same space and the same building element. This approach maintains control over all objects. Thus, individual faces, objects, IFC classes, and spaces can be filtered via regular expressions e. g. in the CFD software. The algorithm ensures consistent triangulation, but mesh quality criteria such as skewness (ratio of internal angles against optimal angle) and aspect ratios cannot be influenced. This is an acceptable procedure since geometry is remeshed anyway when creating surface meshes suitable for CFD purposes.

To export the created objects into the IFC data format, instances of the classes IfcRelSpaceBoundary, IfcSpace and IfcExternalSpatialElement are generated first. In addition to the geometric representation, attributes such as name, description, and the globally unique identifier are then assigned to the entities. For this purpose, a conversion of the geometric classes of OCCT into IFC-based instances is performed using IfcOpenShell. Finally, the entities are linked together to describe the related objects.

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Results

For this paper, the IFC4 *Phantasy office building* of the Institute for Automation and Applied Informatics (IAI) of the Karlsruhe Institute of Technology (KIT) is processed (figure 5). It was chosen because of its good representability and the presence of SBs, which are used for validation of the proposed algorithm.

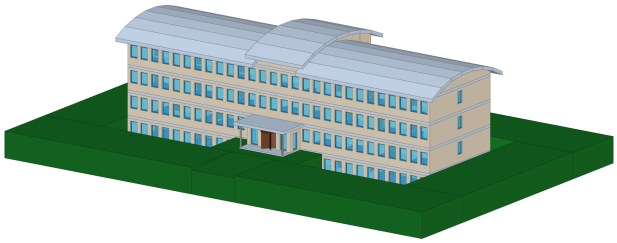
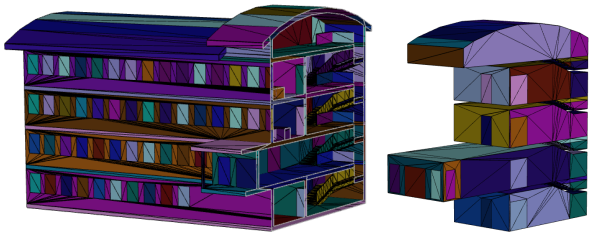


Figure 5: The example IFC building.

STL result

Figure 6 (a) shows a clipped visualization of the resulting STL file. It can be seen that the assignment to the IFC products is preserved despite tessellation and a consistent triangulation exists. However, the triangulation is not optimized for angles or other mesh parameters, which is up to the meshing tool. Figure 6 (b) depicts the staircase zone selected by text filtering based on the naming convention.



(a) clip *(b) staircase zone*

Figure 6: Filtered STL result file.

IFC result

The enriched and exported IFC file contains an IfcExternalSpatialElement, 78 IfcSpaces and almost 2000 IfcRelSpaceBoundaries. The added entities are visualized in figure 8. While subfigure (c) shows the external SBs belonging to IfcSpaces, the subfigures (d) and (e) depict the SBs of the IfcExternalSpatialElement. The typing of the SBs in contact with the ground is not yet complete, since the IfcSite is not yet considered by the algorithm. Unlike the original office model, the enriched model provides SBs of type 2b, creating watertight spaces.

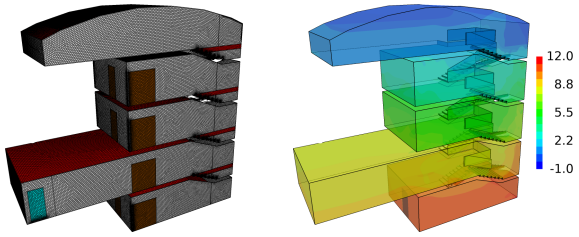
Application

In this section, the generated STL and IFC files will be applied to the simulation process.

CFD

The application of the STL file is shown by means of a steady-state airflow simulation in the staircase, investigating a smoke protection pressure system. The simulation is performed for summer conditions with closed doors. For this purpose, the STL file to be simulated was imported into Fluent Meshing. By naming the surfaces according to zone, IFC class, and related building element, it was possible to delete all building areas except the staircase and to impose the boundary conditions on the desired surfaces in an easy and straight forward manner.

Due to the appropriate preparation, wrapping was not necessary. Instead, stitching was performed, which was based solely on vertex merging. Figure 7 (a) shows the used surface mesh. For volume meshing, the unstructured triangular surface mesh was converted to a hexagonal mesh consisting of 2.8 million cells. The simulation was performed in Fluent using the SST k-ω turbulence model. As one of three flow variables, the resulting air pressure is shown in figure 7 (b).



(a) Surface mesh *(b) Relative pressure in Pa*

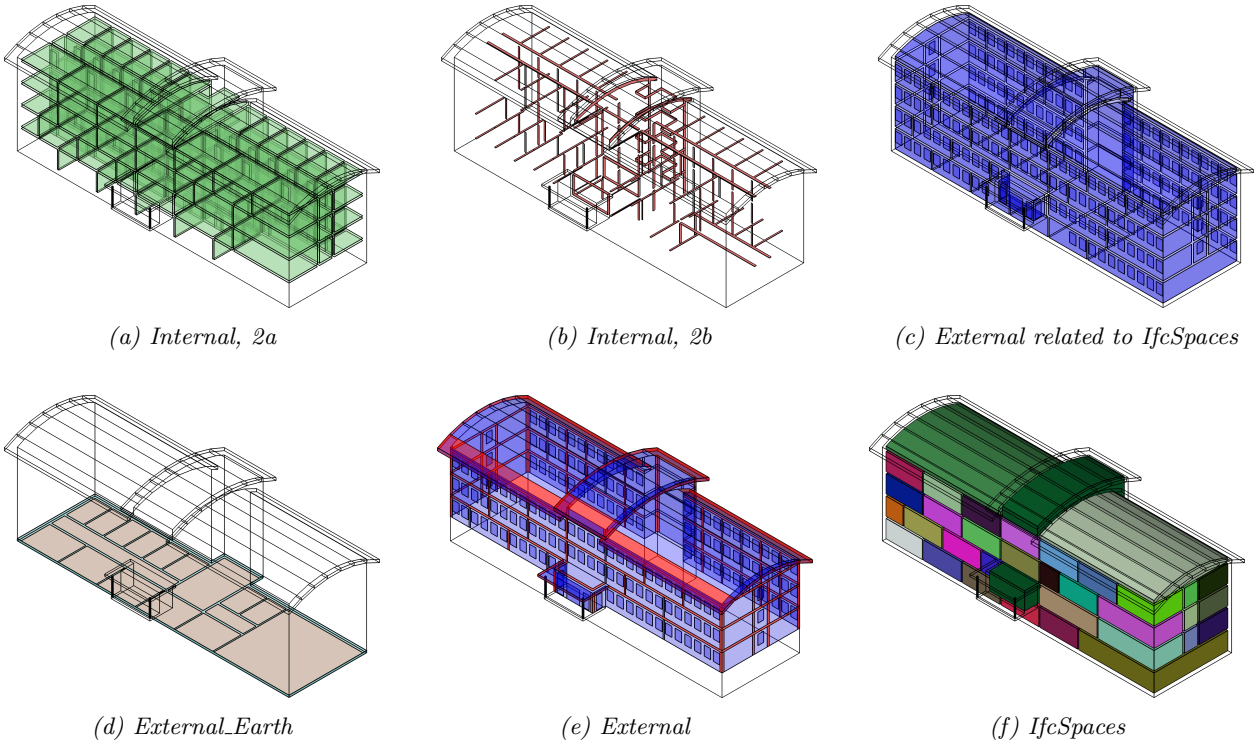
Figure 7: Indoor air flow simulation of a stairway using the exported STL geometry.

BEPS using EnergyPlus

The simulation of the office model enriched with the SBs was performed in EnergyPlus. The needed EnergyPlus Input Data File (IDF) was generated using the BIM2SIM Python toolchain. This toolchain is based on IfcOpenShell and GeomEppy (2016) to convert the SBs objects to IDF surface geometries. According to the IFC file, the IDF surfaces were positioned on the building component surfaces, to correctly represent the thermal mass of the building. SBs of type 2b were used to close the spaces and set to be thermally adiabatic. Each IfcSpace was transferred as a single zone and all non-geometric simulation parameters such as heating and cooling set points, loads, schedules, construction, and materials were chosen according to standardized templates.

Before importing the IFC file into the BIM2SIM toolchain, it was checked for syntactic errors using the IfcCheckingTool and a Python script inspecting the corresponding rules proposed by Ying and Lee (2017). Both inspections reported no issues. Additionally, a visual inspection of the generated IDF in OpenStudio was carried out, examining interior 2a SB

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(a) Internal, 2a *(b) Internal, 2b* *(c) External related to IfcSpaces*

(d) External_Earth *(e) External* *(f) IfcSpaces*

Figure 8: Generated space boundaries highlighted by type.

pairings, surface normals, EnergyPlus boundary conditions, thermal zones and constructions. As shown in figure 9, the geometry is well-defined and all zones appear to be watertight. According to the EnergyPlus error log, no errors attributable to the enriched SBs were reported during the simulation.

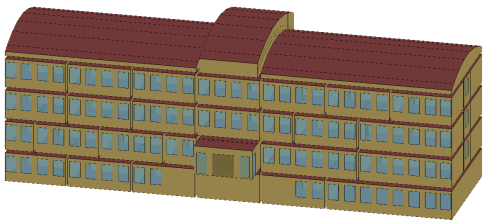


Figure 9: Visualization of the generated IDF based on the enriched IFC.

The positions and areas of the SBs were compared with the SBs of the original office model. The quantitative evaluation showed agreement in the number and position of all zones except the stairwell. This is due to the different zoning, as there are no virtual SBs in the proposed IFC file.

Conclusion

This paper presented an algorithm for generating first and second level space boundaries based on IFC data. The need for this tool was identified during research work and cooperations with industrial partners. Based on two application examples, the functionality of the algorithm and the correctness of the generated data were successfully demonstrated. A significant simplification and acceleration could be observed in the creation of simulation models. The obtained results are a good basis for further improvements of the tool. Future developments may include the support of virtual space boundaries, the increase of the overall robustness, and the subdivision of faces in contact with the site. This tool contributes to an improved BIM work process for building simulations. This includes investigations in which a BEPS as well as a CFD simulation must be performed. By supporting both simulation geometries, model setup can be simplified and boundary conditions can be exchanged more easily between the simulations.

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